Loss functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Loss functions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Parameters and architecture

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Composing layers

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dot products and weighted sums

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A loss assigns a numerical penalty to a **prediction** compared with a **target**. For this regression lesson, **residual** e = **prediction** - **target**, and **squared error** is e squared.

Squaring makes both positive and negative **residual**s contribute nonnegative penalties. We use ordinary **mean squared error** without a one-half factor.

Example · Target · Prediction · Residual · Squared error

Example: A

Target: 1

Prediction: 2

Residual: 1

> **Squared error:** 1

> **Example:** B
>
> **Target:** 3
>
> **Prediction:** 2
>
> **Residual:** -1
>
> **Squared error:** 1

> **Example:** C
>
> **Target:** 2
>
> **Prediction:** 4
>
> **Residual:** 2
>
> **Squared error:** 4

The **mean squared error**, or MSE, is (1 + 1 + 4) / 3 = 2. It summarises these three examples, not every possible future **input**. If **target**s use cm, **residual**s use cm and MSE uses square cm.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Loss functions trace and record the first step you can explain without help.

Use W2: Expose cancellation for Loss functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Loss functions explanation against the supplied facts before discussing it.

Use the worked case for Loss functions. Proposed explanation: “Average **residual** is the loss”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Compare an unfamiliar evaluation independently

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Calculate a loss

> Calculate a loss. Apply the supplied Loss functions facts and keep your reasoning.

For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Expose cancellation

> Expose cancellation. Apply the supplied Loss functions facts and keep your reasoning.

Targets [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

My reasoning and evidence

W3 Check the denominator

> Check the denominator. Apply the supplied Loss functions facts and keep your reasoning.

A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

My reasoning and evidence

W4 Compare an unfamiliar evaluation independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Compare an unfamiliar evaluation independently. Apply the supplied Loss functions facts and keep your reasoning.

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

> **My reasoning and evidence**

W5 Protect final testing

> Protect final testing. Apply the supplied Loss functions facts and keep your reasoning.

> A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Loss functions to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled loss functions **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate per-example **squared error** and **mean squared error**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Loss functions

> Compare candidate losses with the same targets and denominator. Explain the labelled step before checking the changed condition.

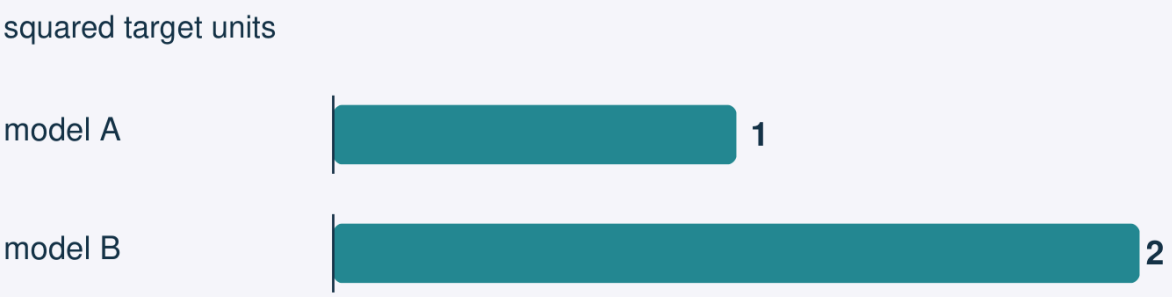
Compare candidate losses with the same **target**s and denominator.

Targets [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/8

MSE comparison on shared targets



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/8

Loss functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. A squared errors

A residuals [-1, 1] give squares [1, 1], MSE 1.



2. B squared errors

B residuals [0, 2] give squares [0, 4], MSE 2.



3. Compare objective

A wins under MSE; both MAEs equal 1.

CHANGE ONE THING

Choose MAE instead of MSE as the sole criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare candidate losses with the same **target**s and denominator.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A wins under MSE; both MAEs equal 1.

1. Compare objective

2. A **squared error**s

3. B **squared error**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare objective A wins under MSE; both MAEs equal 1. For this loss calculation, the signed error is **prediction** minus **target**.

MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case. Squaring makes the larger individual error count more.

State the objective before selecting a model, because these two objectives give different comparisons.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Choose MAE instead of MSE as the sole criterion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. A and B tie at MAE 1.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A and B tie at MAE 1. **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Repair a missing building block

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

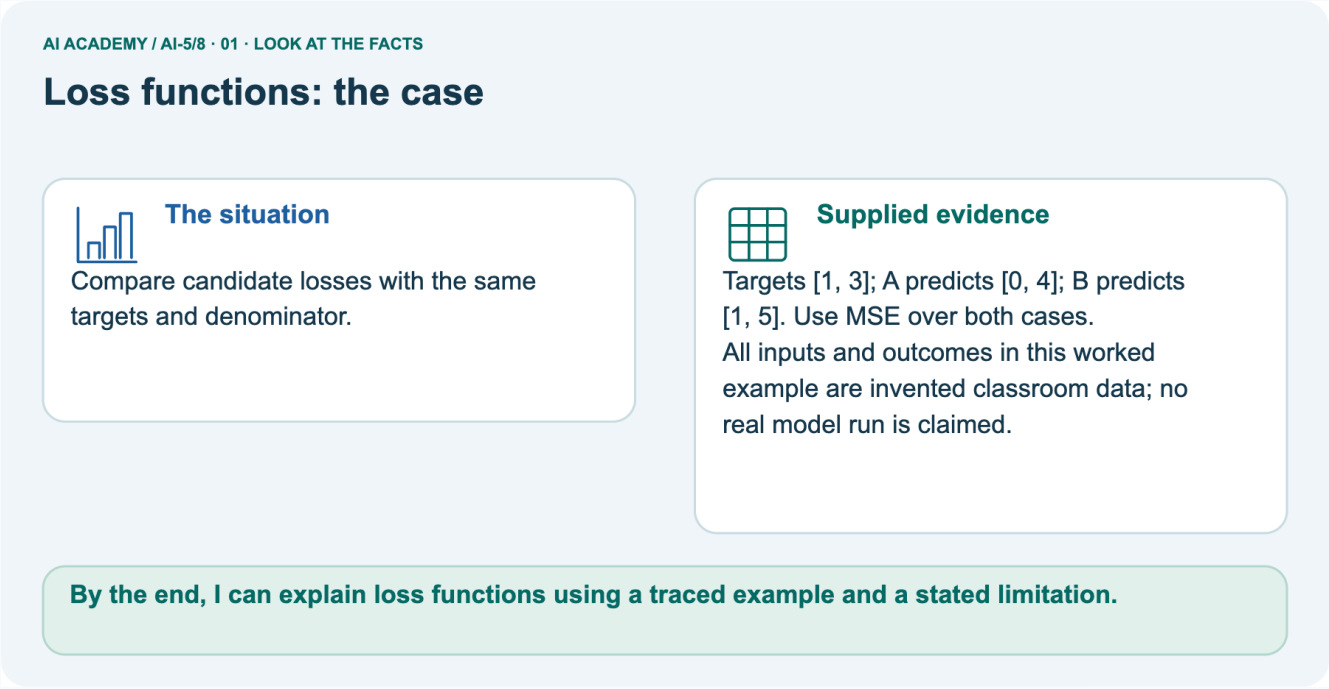
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Loss functions picture from its labels, then record a relevant limit.

01. Loss functions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

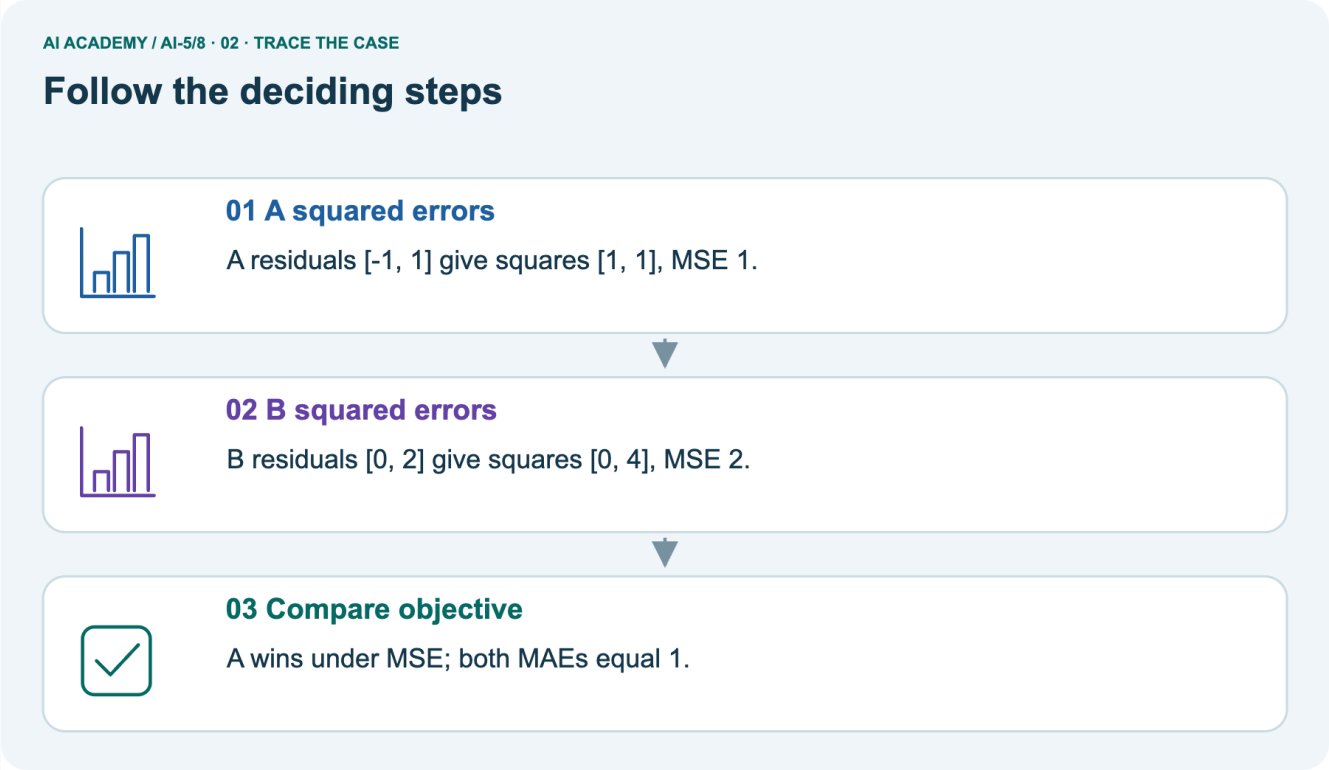
Compare candidate losses with the same **target**s and denominator. Supplied facts: **Target**s [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



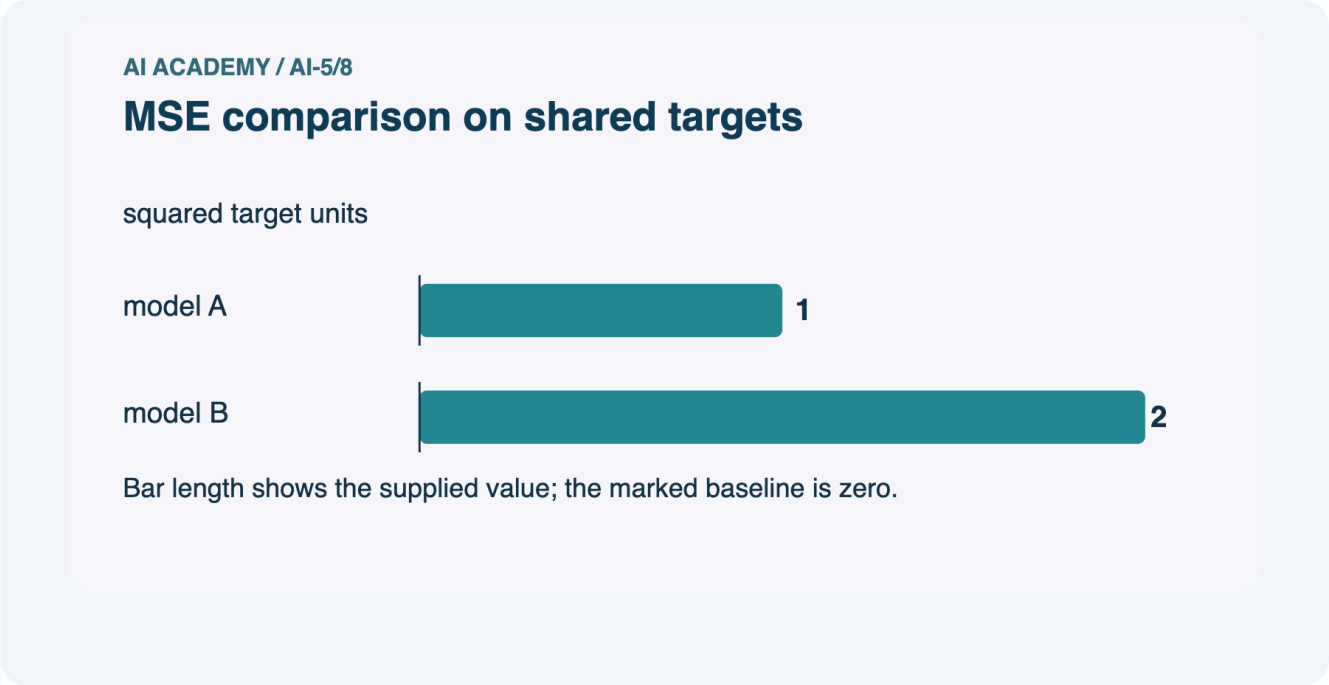
Follow the labelled picture with the matching explanation. Open the vector diagram

A **squared error**s: A **residual**s [-1, 1] give squares [1, 1], MSE 1. → B **squared error**s: B **residual**s [0, 2] give squares [0, 4], MSE 2. → Compare objective: A wins under MSE; both MAEs equal 1.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. MSE comparison on shared targets



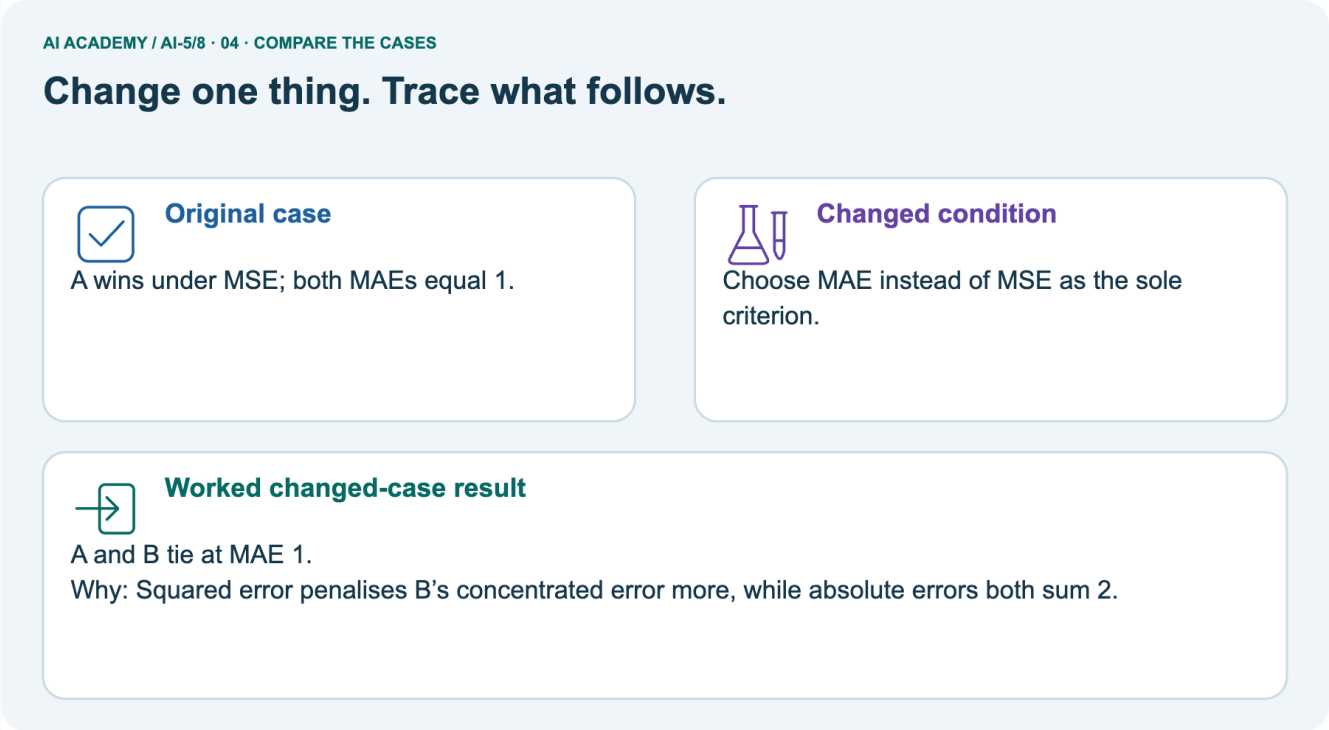
Follow the labelled picture with the matching explanation. Open the vector diagram

MSE comparison on shared **target**s. Units: squared **target** units. model A: 1. model B: 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

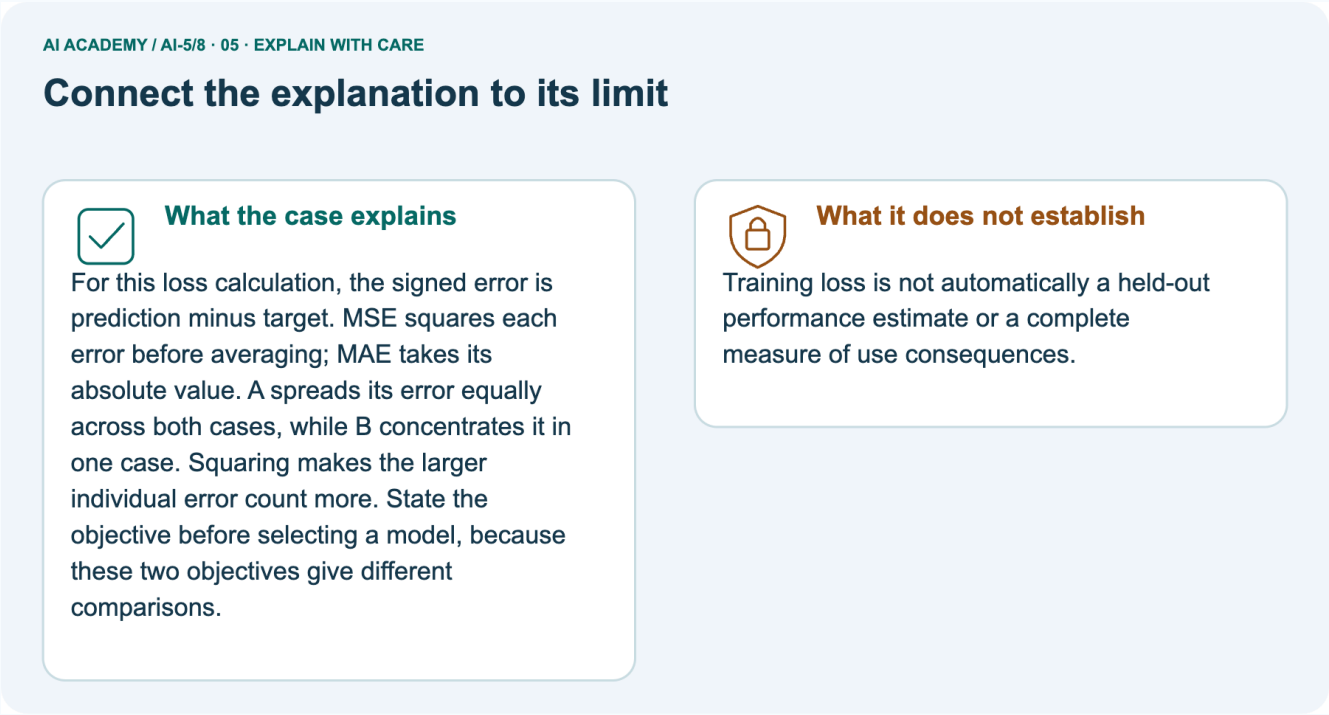
Original case: A wins under MSE; both MAEs equal 1. Changed condition: Choose MAE instead of MSE as the sole criterion. Expected result: A and B tie at MAE 1.

Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

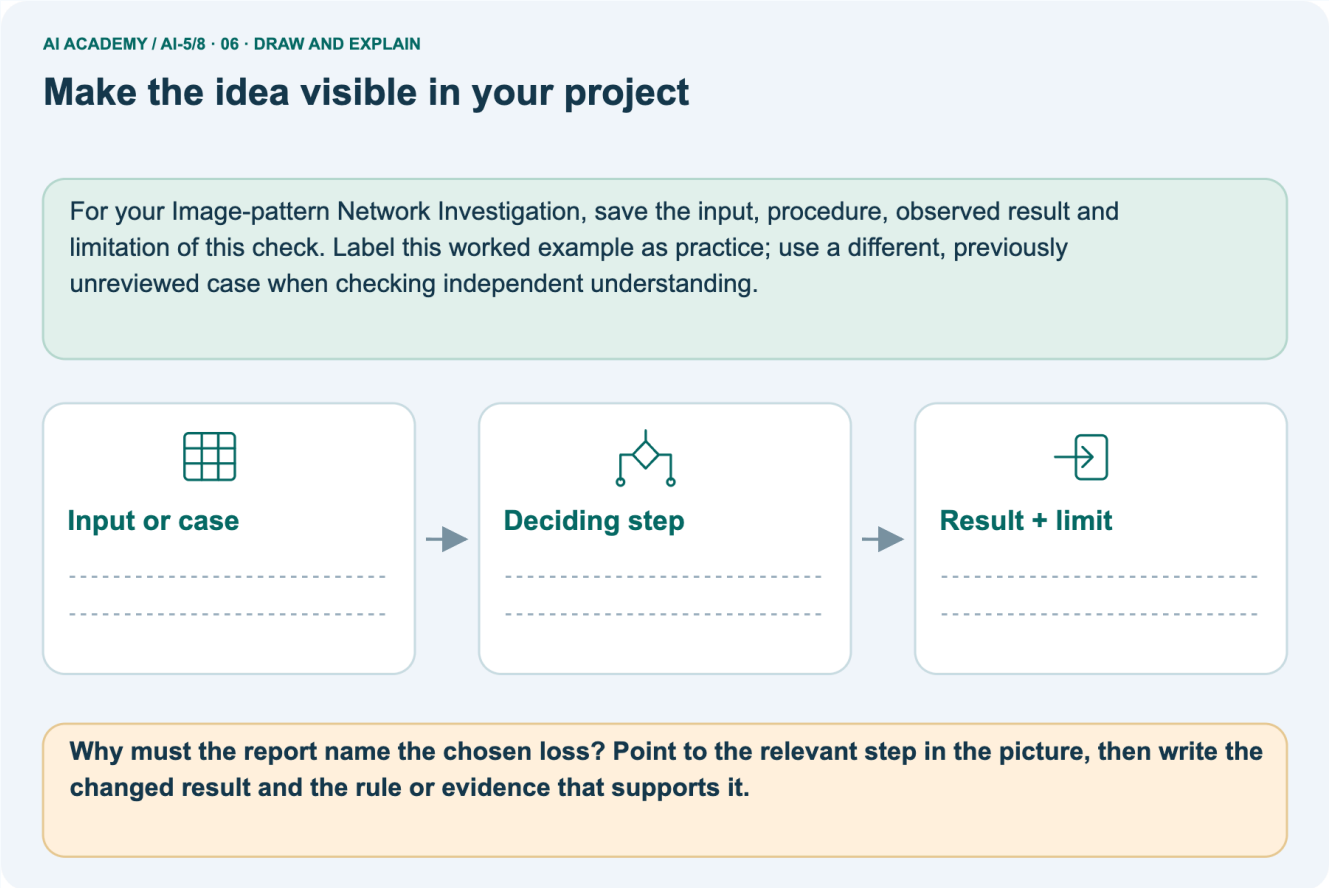
Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Limit: **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.